

Omid Rohanian

ASSISTANT PROFESSOR · UNIVERSITY OF COPENHAGEN · NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING

University of Copenhagen, Copenhagen, Denmark

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Profile

Assistant Professor at the University of Copenhagen working on natural language processing and machine learning for scientific, clinical, and social data. My research focuses on biomedical and clinical NLP, efficient language models, domain adaptation, and the evaluation of language technologies in settings where reliability, interpretability, and responsible use matter. I have developed compact transformer models and open biomedical NLP resources, and my current work extends these methods to text, registry data, and other complex datasets in social data science.

Work Experience

University of Copenhagen

Copenhagen, Denmark

ASSISTANT PROFESSOR

Jun 2026 - Present

- Conduct research in natural language processing and machine learning for scientific, clinical, and social data.
- Develop methods for analysing text, registry data, and other complex datasets in social data science.
- Continue work on biomedical and clinical NLP, efficient language models, and evaluation of language technologies in high-stakes domains.

University of Oxford

Oxford, United Kingdom

VISITING RESEARCHER, DEPARTMENT OF ENGINEERING SCIENCE

Nov 2025 - Present

- Continue collaborations on efficient biomedical and clinical NLP, large clinical datasets, open resources, and publication-led research.
- Contribute to evaluation and privacy-aware workflows for clinical text processing and responsible model deployment.

QA Ltd

London, United Kingdom

SENIOR TECHNICAL LEARNING SPECIALIST

Nov 2025 - Apr 2026

- Designed and delivered hands-on professional training and mentoring in generative AI, NLP, and machine learning for public sector and industry audiences.
- Developed curricula, practical labs, and teaching materials tailored to mixed-ability cohorts and organisational needs.
- Advised teams on applying LLM and NLP methods to real-world problems, with emphasis on evaluation, limitations, privacy, and responsible use.
- BCS Certified Trainer for the *BCS Essentials Certificate in Artificial Intelligence*; TAP-qualified and applied structured instructional design to improve learning outcomes.

University of Oxford

Oxford, United Kingdom

SENIOR RESEARCH ASSOCIATE, DEPARTMENT OF ENGINEERING SCIENCE

Jul 2023 - Nov 2025

- Led and contributed to research in biomedical and clinical NLP, with a focus on efficient transformer models, model distillation, and domain-adapted language technologies.
- Developed and released open biomedical language models and resources that have been used by NLP and biomedical research communities.
- Contributed to the GLOPID-R Pandemic PACT project, applying language technologies to annotate and monitor global pathogen research activities.
- Worked with collaborators across clinical and biomedical datasets, supporting publication-led research and reproducible model development.

University of Oxford

Oxford, United Kingdom

POSTDOCTORAL RESEARCHER, DEPARTMENT OF ENGINEERING SCIENCE

Oct 2020 - Dec 2023

- Contributed to the CURIAL-Rapide project, developing machine learning methods for rapid COVID-19 screening using routine blood test data.
- Managed data preprocessing, modelling, and analysis workflows for clinical machine learning research.
- Developed NLP and machine learning methods for biomedical and clinical text processing in collaboration with clinical and engineering researchers.

Education

University of Wolverhampton

Wolverhampton, United Kingdom

PHD IN NATURAL LANGUAGE PROCESSING

2017 - 2020

- **Dissertation:** "Computational Treatment of Non-literal Language"
- Research on computational modelling of figurative language, contextualised representation learning, and linguistically informed neural architectures.

Sharif University of Technology

Tehran, Iran

MSc IN NATURAL LANGUAGE PROCESSING

2013 - 2016

- **Dissertation:** "Natural Language Generation From Visual Input"
- Developed a multimodal natural language generation system for translating visual information into textual descriptions.

Honours & Awards

2025	Trelis AI Micro-Grant , Trelis Research, competitive micro-grant programme	Global
2024	Best Paper in NLP , International Medical Informatics Association (IMIA) Yearbook, global selection	Global
2022	Global Talent Endorsement , UK Research and Innovation	United Kingdom
2018	Ranked 1st , Automatic Identification of Verbal Multiword Expressions, LAW-MWE-CxG Shared Task	Santa Fe, USA
2018	Ranked 3rd , SemEval 2018 Task 3: Irony Detection in English Tweets	New Orleans, USA
2018	Ranked 4th (joint) , SemEval 2018 Task 10: Capturing Discriminative Attributes	New Orleans, USA
2017	Best Paper Award , Recent Advances in Natural Language Processing (RANLP)	Varna, Bulgaria
2017	Best Paper by a Young Researcher Award , EUROPHRAS	Las Vegas, USA
2016	Full PhD Studentship , University of Wolverhampton	United Kingdom

Selected Publications¹

- Nooralahzadeh, F., **Rohanian, O.**, Zhang, Y., Fürst, J., Stockinger, K. *Arbitration Failure, Not Perceptual Blindness: How Vision-Language Models Resolve Visual-Linguistic Conflicts*. (2026 - link to preprint, arXiv, 2604.09364).
- Nouriborji, M., Rohanian, M., **Rohanian, O.** *Improving Recursive Transformers with Mixture of LoRAs*. (2025 - link to preprint, arXiv, 2512.12880, cs.LG).
- Liu, F., Li, Z., Zhou, H., Yin, Q., Yang, J., Tang, X., Luo, C., Zeng, M., Jiang, H., Gao, Y., Nigam, P., Nag, S., Yin, B., Hua, Y., Zhou, X., **Rohanian, O.**, Thakur, A., Clifton, L., Clifton, D. A. *Large Language Models Are Poor Clinical Decision-Makers: A Comprehensive Benchmark*. (2024 - link to publication, Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 13696–13710).
- Rohanian, O.**, Nouriborji, M., Seminog, O., Furst, R., Mendy, T., Levanita, S., Kadri-Alabi, Z., Jabin, N., Toale, D., Humphreys, G., Antonio, E., Bucher, A., Norton, A., Clifton, D. A. *Rapid Biomedical Research Classification: The Pandemic PACT Advanced Categorisation Engine*. (2024 - link to preprint, arXiv, 2407.10086, cs.CL).
- Rohanian, O.**, Nouriborji, M., Kouchaki, S., Nooralahzadeh, F., Clifton, L., Clifton, D. A. *Exploring the Effectiveness of Instruction Tuning in Biomedical Language Processing*. (2024 - link to publication, Artificial Intelligence in Medicine, 158, 103007).
- Taylor, N., Ghose, U., **Rohanian, O.**, Nouriborji, M., Kormilitzin, A., Clifton, D. A., Nevado-Holgado, A. *Efficiency at Scale: Investigating the Performance of Diminutive Language Models in Clinical Tasks*. (2024 - link to publication, Artificial Intelligence in Medicine, 157, 103002).
- Seminog, O., Furst, R., Mendy, T., **Rohanian, O.**, Levanita, S., Kadri-Alabi, Z., Jabin, N., Humphreys, G., Antonio, E., Bucher, A., Norton, A. *A Protocol for a Living Mapping Review of Global Research Funding for Infectious Diseases with a Pandemic Potential–PANDEMIC PACT*. (2024 - link to paper, Wellcome Open Research, Vol. 9, Issue 156, pp. 156).
- Chauhan, V. K., Thakur, A., O'Donoghue, O., **Rohanian, O.**, Molaei, S., Clifton, D. A. *Continuous Patient State Attention Model for Addressing Irregularity in Electronic Health Records*. (2024 - link to paper, BMC Medical Informatics and Decision Making, Vol. 24, Issue 1, pp. 117).
- Rohanian, O.**, Nouriborji, M., Jauncey, H., Kouchaki, S., Nooralahzadeh, F., Clifton, L., Merson, L., Clifton, D. A., ISARIC Clinical Characterisation Group. *Lightweight Transformers for Clinical Natural Language Processing*. (2023 - link to paper, Natural Language Engineering, pp. 1–28).
- Rohanian, O.**, Nouriborji, M., Kouchaki, S., Clifton, D. A. *On the Effectiveness of Compact Biomedical Transformers*. (2023 - Bioinformatics, Vol. 39, No. 3, Article btad103, DOI: 10.1093/bioinformatics/btad103).
- Rohanian, M., Nooralahzadeh, F., **Rohanian, O.**, Clifton, D., Krauthammer, M. *Disfluent Cues for Enhanced Speech Understanding in Large Language Models*. In Findings of the Association for Computational Linguistics: EMNLP 2023, pages 3676–3684, December 6–10, 2023. (link to paper).
- Rohanian, O.**, Kouchaki, S., Soltan, A. A. S., Yang, J., Rohanian, M., Yang, Y., Clifton, D. A. *Privacy-Aware Early Detection of COVID-19 Through Adversarial Training*. (2023 - Journal of Biomedical and Health Informatics, Vol. 27, No. 3, pp. 1249–1258, DOI: 10.1109/JBHI.2022.3230663) (link to paper).

¹For the full list of publications, please visit my Google Scholar page.

13. Nouriborji, M., **Rohanian, O.**, Kouchaki, S., Clifton, D. A. *MiniALBERT: Model Distillation via Parameter-Efficient Recursive Transformers*. (2023 - link to paper, Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics, pp. 1161–1173, Dubrovnik, Croatia).
14. **Rohanian, O.**, Jauncey, H. et al. *Using Bottleneck Adapters to Identify Cancer in Clinical Notes under Low-Resource Constraints*. In Proceedings of the 22nd Workshop on Biomedical Natural Language Processing and BioNLP Shared Tasks (BioNLP 2023). (link to paper).
15. Nouriborji, M., **Rohanian, O.**, Clifton, D. A. (2022) *Nowruz at SemEval-2022 Task 7: Tackling Cloze Tests with Transformers and Ordinal Regression*. In Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022). (link to paper).
16. Soltan, A. A. S., Yang, J., Pattanshetty, R., Novak, A., Yang, Y., **Rohanian, O.**, et al. *Real-world evaluation of AI-driven COVID-19 triage for emergency admissions: External validation & operational assessment of lab-free and high-throughput screening solutions*. (2022 - *Lancet Digital Health*). (link to paper).
17. **Rohanian, O.**, Rei, M., Taslimipoor, S., Ha, L. A. (2020) *Verbal Multiword Expressions for Identification of Metaphor*. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL 2020). (link to paper).
18. **Rohanian, O.**, Taslimipoor, S., Kouchaki, S., Ha, L. A., Mitkov, R. *Bridging the Gap: Attending to Discontinuity in Identification of Multiword Expressions*. In Proceedings of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL 2019). (link to paper).

Selected Research Projects and Open Resources

NLPIE Research: Biomedical Language Models and Open NLP Resources

NLPIE Research

CO-DEVELOPER AND RESEARCH LEAD

2022 - Present

- Developed and released biomedical and clinical language models through NLPIE Research, spanning compact encoder models such as CompactBioBERT and instruction-tuned generative models such as Llama2-MedTuned 7B and 13B.
- Built open resources for biomedical NLP, including instruction-tuned Llama 2 models trained on approximately 200K task-focused biomedical and clinical instruction samples.
- Applied knowledge distillation, continual pre-training, instruction tuning, recursive transformer architectures, and parameter-efficient adaptation to improve the efficiency, usability, and accessibility of biomedical language technologies.

ModernALBERT and Mixture of LoRAs for Recursive Transformers

NLPIE Research

CO-DEVELOPER AND MAINTAINER

2025 - Present

- Contributed to ModernALBERT, a family of modernised recursive transformer models designed to improve the expressivity and efficiency of compact language models.
- Explored Mixture of LoRAs as a lightweight conditional-computation mechanism for restoring layer-wise expressivity under aggressive parameter sharing.
- Released model variants and associated resources through NLPIE Research as part of an open compact-model research programme.

Pandemic PACT Advanced Categorisation Engine

University of Oxford / GLOPID-R

AI AND MACHINE LEARNING LEAD

Feb 2023 - Nov 2025

- Led the AI and machine learning work for automated classification and annotation of global infectious disease research records.
- Developed NLP and large-language-model methods to support research monitoring, trend identification, and evidence mapping for pandemic preparedness.

ISARIC Clinical Notes Project

University of Oxford

RESEARCH LEAD

Dec 2021 - Oct 2024

- Led NLP research on international clinical notes, focusing on disease mention detection, clinical concept extraction, and linkage to biomedical knowledge resources.
- Worked with heterogeneous clinical text data to support reusable methods for clinical information extraction across settings and datasets.

Privacy-Aware Clinical Machine Learning

University of Oxford

RESEARCH CONTRIBUTOR

Jul 2021 - Mar 2022

- Investigated privacy-aware machine learning methods for clinical and tabular data, including adversarial approaches to reducing sensitive-information leakage.
- Contributed to modelling workflows for early clinical risk detection using routinely collected healthcare data.

Teaching and Academic Service

Professional AI, Data Science, and NLP Training, QA Ltd

London, United Kingdom

COURSE DESIGNER AND INSTRUCTOR

Nov 2025 - Apr 2026

- Designed and delivered professional courses in data science, generative AI, NLP, and LLM-based text analysis for industry and public-sector learners.
- Developed practical teaching materials, labs, and applied examples, including domain-specific material for healthcare and biomedical data analysis.

Introduction to Biomedical NLP, AI for Efficient Healthcare Systems, University of Oxford

Oxford, United Kingdom

LECTURER AND LAB INSTRUCTOR

2024

- Delivered lecture and lab teaching on biomedical NLP, with emphasis on clinical text, healthcare data, and practical NLP workflows.

Introduction to Natural Language Processing, EPSRC CDT in Health Data Science, University of Oxford

Oxford, United Kingdom

INVITED LECTURER

Nov 2023 & Nov 2024

- Delivered annual invited lectures on foundational NLP concepts for doctoral students in health data science.

Major NLP Conferences and Workshops

PROGRAMME COMMITTEE MEMBER AND REVIEWER

2018 - 2026

- Reviewed submissions for leading NLP venues, including ACL, EMNLP, NAACL, EACL, and associated *ACL conferences and workshops.

Cambridge Journal of Natural Language Engineering

Wolverhampton, United Kingdom

EDITORIAL ASSISTANT

Sep 2016 - Sep 2020

- Supported editorial workflows for a leading journal in natural language processing and computational linguistics.

RANLP 2019 Summer School on Deep Learning in Natural Language Processing

Varna, Bulgaria

CO-ORGANISER AND TUTOR

Aug 2019

- Co-organised teaching activities and delivered a hands-on PyTorch session on sequence labelling and tagging.

Machine Learning for NLP and Computational Linguistics, University of Wolverhampton

Wolverhampton, United Kingdom

TEACHING ASSISTANT

2018 - 2020

- Assisted with master's-level teaching in computational linguistics and machine learning for NLP.

Skills

Research Methods	Natural language processing, machine learning, deep learning, biomedical and clinical NLP, text-as-data, model evaluation, error analysis, reproducible research.
Language Models	Transformer models, compact and efficient language models, knowledge distillation, domain adaptation, instruction tuning, parameter-efficient fine-tuning, prompt-based methods.
Biomedical and Clinical AI	Clinical text processing, biomedical information extraction, medical NLI, relation extraction, named entity recognition, privacy-aware clinical machine learning.
Programming and Tools	Python, PyTorch, Hugging Face Transformers, PEFT, Scikit-learn, Pandas, NumPy, Jupyter, Git, Bash, SQL, vector databases.
Teaching and Communication	Academic writing, research communication, curriculum design, technical mentoring, professional training, interdisciplinary collaboration.
Languages	Persian (native), English (full professional proficiency), Spanish (intermediate; B1).

Professional Certifications

2026	Certificate of Competency: Fundamentals of Accelerated Data Science , NVIDIA	<i>United Kingdom</i>
2025	TAP Certificate for Trainers , Training Accreditation Programme	<i>United Kingdom</i>
2025	BCS Essentials Certificate in Artificial Intelligence , British Computer Society, Certified Trainer	<i>United Kingdom</i>